

Aashish Sheshadri

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PROFILE

Twelve years building the layer that lets unreliable automated decision-makers act in production safely. Today that layer is the agentic SDLC platform, where no agent holds a write path. Before it: **graduated autonomy**, now the company-wide standard for how much autonomy an ML system is granted, and a hierarchical RL policy for cluster capacity where the hard part was reward specification, not policy architecture. Individual contributor throughout — the standards I set were adopted by orgs I do not run.

5 peer-reviewed publications, 362 citations across all work (Google Scholar, Aug 2026), 1 issued patent and 1 pending as lead inventor, and AAAI HCOMP's Test of Time Award for SQUARE.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript; Go, C, SQL

Frameworks: PyTorch, TensorFlow, Kubernetes, GCP/AWS/Azure, Spark, Ray, Jupyter

Domains: Agentic Systems & Tool-Use Harnesses, Reward & Curriculum Design, Evaluation Methodology, Reinforcement Learning, Deployment Safety & Oversight, ML Infrastructure, Time-Series Forecasting

EXPERIENCE

PayPal Inc, San Jose, CA

June 2014 – Present

Machine Learning Architect (Sr. MTS)

Feb 2023 – Present

Set ML and infrastructure strategy across cloud platform, data, and developer productivity. Decide which problems get an ML solution vs. a heuristic, select modeling approaches, and govern production rollouts across ~1,550 applications.

Agentic SDLC — Infra, Application, and Data

- Built the multi-agent platform that carries an engineer's stated intent to production across the infrastructure, application and data lifecycles — provisioning, onboarding, per-change environments, functional-test gates, canary promotion. Specialist LLM agents behind one orchestrator, sharing memory, policy and a skill catalog rather than each rebuilding them. In production since August 2026; too early for adoption numbers.
- Set the safety envelope: no agent holds a write path, and every mutation is executed by deterministic code acting on an agent's proposal. Authority is budgeted by blast radius — an open-ended tool loop for the agent furthest from production, a short schema-validated pipeline for the one that touches it.
- Made promotion a code check rather than an instruction: an evidence-first evaluator scores acceptance criteria and a state transition keyed on that verdict holds the change in draft until it passes.

ML systems and deployment governance

- Designed the ML deployment framework — canary evaluation, shadow mode, **graduated autonomy** — now the company-wide production standard governing how much autonomy an ML system is granted. Introduced as a Staff engineer.
- Defined what evidence a system must produce before its autonomy widens, so the question is answered by a gate rather than a review meeting.
- Set autoscaling and capacity strategy, and built the hierarchical RL policy for cluster capacity allocation across ~1,550 applications (~100k pods), acting on the demand forecasts I had built for the same fleet. The hard part was reward function and curriculum engineering, not the policy architecture.

- Found and closed reward-specification failures under optimization pressure — cases where the policy optimized the written objective into an outcome I had not intended. Optimizer-found, not model-reasoned: the policy is small and non-linguistic and cannot represent its own evaluation.
- Ongoing collaboration with IBM Quantum on variational algorithms for risk modeling (published in *Quantum*, 2023).

Staff Machine Learning Engineer (MTS 2)

Mar 2020 – Feb 2023

- Led ML-driven predictive autoscaling for ~1,500 applications, working toward five-nines availability while cutting compute spend ~25%.
- Carried the sequence-forecasting work from infrastructure signals to demand: time-series models with confidence-calibrated predictions for ~1,500 applications, handling distribution shift across seasonal, promotional, and organic traffic patterns. Those forecasts became the autoscaler's input signal.
- Developed sequence classification models using BERT for real-time threat detection in information security — security-critical inference where false negatives are expensive.

Senior Machine Learning Engineer (MTS 1)

Mar 2018 – Feb 2020

- Applied sequence-to-sequence models (RNN/LSTM) to CPU and memory forecasting, replacing hand-tuned static alert thresholds with likelihood-based outlier detection, and scaled it across the ~350,000-signal monitoring surface. Presented the approach at QCon.ai 2019 — infoq.com/presentations/journey-99999
- Built the enterprise ML platform: multi-tenant GPU scheduling with prioritized access, remote kernel orchestration, and isolated environments from notebook through production serving. Teams shipped without competing for shared compute.
- Wrote the parameterization layer PPExtensions' `run_pipeline` magic delegates to — the AST-level substitution that type-checks and injects values into each notebook's parameter cell, which is what hands state between pipeline stages. Plus Airflow scheduling and the CSV, logging and config layers (github.com/paypal/PPExtensions).

Senior Research Engineer

Mar 2016 – Feb 2018

- Co-authored the FASH64 specification with Doug Crockford (JSON inventor): a fast non-cryptographic hash, validated by statistical dispersion analysis and empirical collision-rate testing (crockford.com/fash64.pdf).
- Designed and open-sourced the **Seif Protocol**, an ECC-512 handshake over Node.js (github.com/paypal/seif-protocol).
- Research in probabilistic analysis and randomness validation — mathematical foundations later applied to distribution modeling and uncertainty quantification in ML systems.

Research Engineer

Jun 2014 – Feb 2016

- Built the cryptographic foundation under Doug Crockford: **seifnode**, a Node.js addon over ISAAC RNG, ECC, AES-GCM and SHA3-256, and the C++ entropy-mining CSPRNG beneath it. 543 stars, MIT (github.com/paypal/seifnode). Presented at OSCON, 2015 and 2016.

The University of Texas at Austin, Austin, TX

Sept 2012 – May 2014

Graduate Research Assistant

- Research in IR evaluation and human computation: aggregating noisy, disagreeing human judgments into reliable evaluation signal, and hybrid expert/non-expert judging for building test collections affordably — the evaluation-quality problems that recur today in annotation aggregation and model-graded evaluation.
- Built and open-sourced the SQUARE benchmark (MIT) — a common harness for comparing crowd-consensus algorithms (majority vote, GLAD, CUBAM, Dawid-Skene) under supervised, semi-supervised, and unsupervised regimes: github.com/utir/square, and a second-generation Java implementation adding n-fold evaluation and a worker-simulation framework at github.com/utir/square-2.0.

Teaching Assistant

Sept 2012 – Dec 2012

- Programming Languages (undergraduate course)

Carnegie Mellon University, Pittsburgh, PA

Sept 2011 – May 2012

Research Associate

- LiDAR-camera calibration for mapping and localization. Visual odometry and orthographic views for global localization of planetary rovers. Contributed to Astrobotic's Google Lunar XPRIZE effort.

EDUCATION

The University of Texas at Austin — M.S. Computer Science, GPA 3.90

Aug 2012 – May 2014

Computer Vision & IR Thesis Track

Carnegie Mellon University — Audited Courses in Computer Vision, Robotics & Perception

Aug 2011 – May 2012

PES Institute of Technology, Bangalore — B.S. Electronics & Communication, GPA 8.40, Honors

Jun 2007 – Aug 2011

PUBLICATIONS

- Variational quantum algorithm for unconstrained black box binary optimization: Application to feature selection — *Quantum*, Jan 2023
- Mix and match: collaborative expert-crowd judging for building test collections accurately and affordably — *DESIRES*, Aug 2018
- A collaborative approach to IR evaluation (Masters Thesis) — *UT Digital Repository*, May 2014
- SQUARE: A Benchmark for Research on Computing Crowd Consensus (First Author, **Test of Time Award, Oct 2024**) — *AAAI HCOMP*, Nov 2013. Code: github.com/utir/square
- Position Estimation by Registration to Planetary Terrain (First Author) — *IEEE MFI*, Sept 2012
- Complementary Flyover and Rover Sensing for Superior Modeling of Planetary Features — *FSR*, July 2012

PATENTS

- **Session Management System** (Lead Inventor) — US11818219, Issued Nov 14, 2023
- **Dynamic Autoscaling of Server Resources Using Intelligent Demand Analytic Systems** (Lead Inventor) — application US 17/826,071, filed May 26, 2022 (pending)

INVITED SPEAKER

QCon, Strata, MLConf, Deep Learning World, RE•WORK Enterprise AI Summit, OSCON, Fluent

Recorded: *On a Deep Journey towards Five Nines* — QCon.ai 2019, infoq.com/presentations/journey-99999

PEER REVIEWER

Information Processing & Management (IF 7.4), Artificial Intelligence (IF 5.1)